

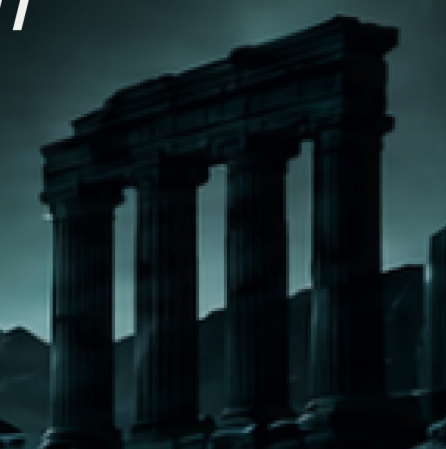


PROJECT MAIEUTIC / 學問

From Inquiry to Software

The CityUHK Initiative on
Specification-Driven & Agentic Computing Education

"The art of bringing
latent knowledge
into light through
questioning."



FROM FIRST YEAR ONWARD,
STUDENTS LEARN TO
THINK, SPECIFY, AND
BUILD WITH AI AGENTS.

HIDDEN IN SOUND

Audio Steganography in a Python Package

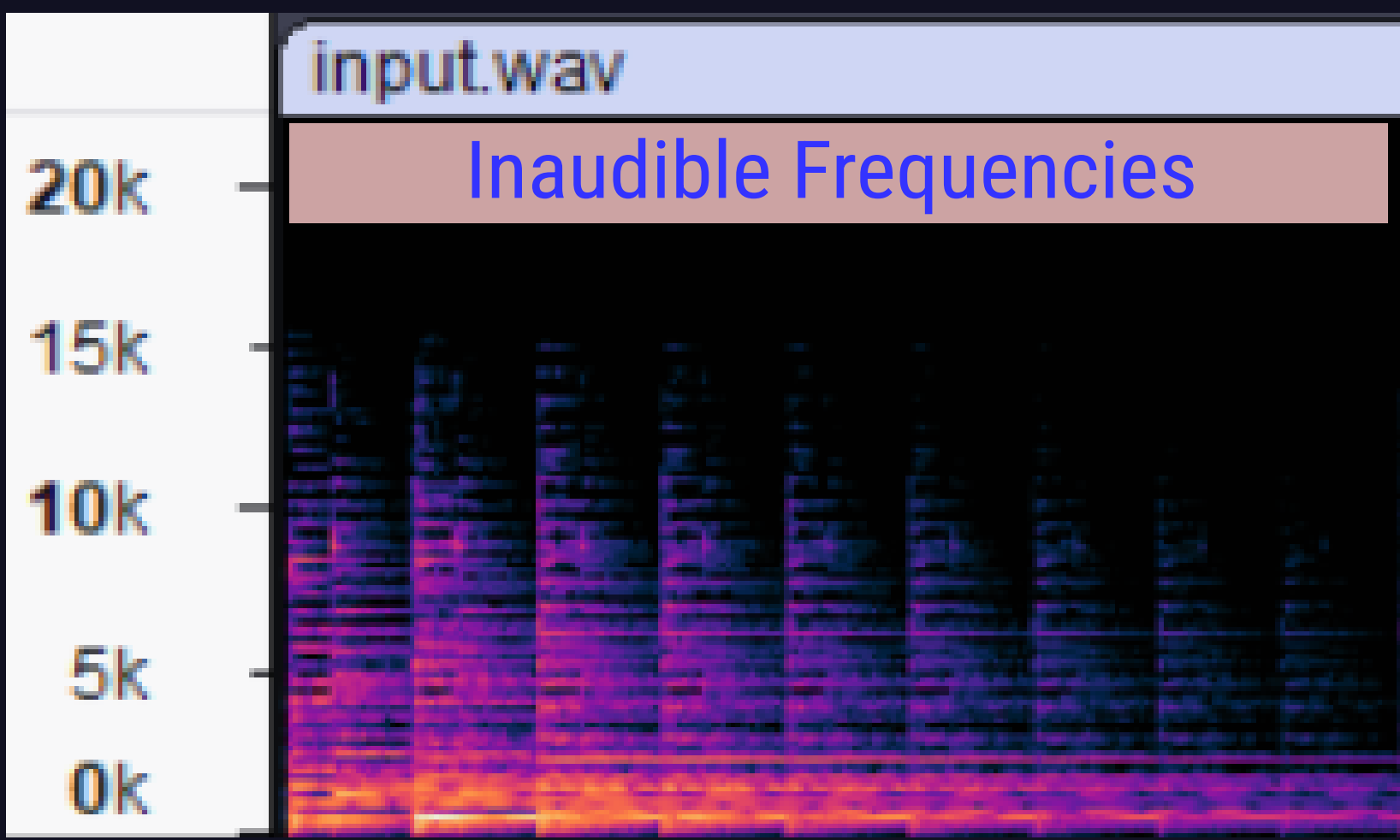
Ultrasonic-img-steg allows storing data in audio files (and extracting the data later), as well as embedding images into the audio's spectrogram, all inside the 19-22 kHz range so most people don't hear a difference!

★ **FIRST-YEAR PROJECT**

Built by a Year-1
CS Dept Student

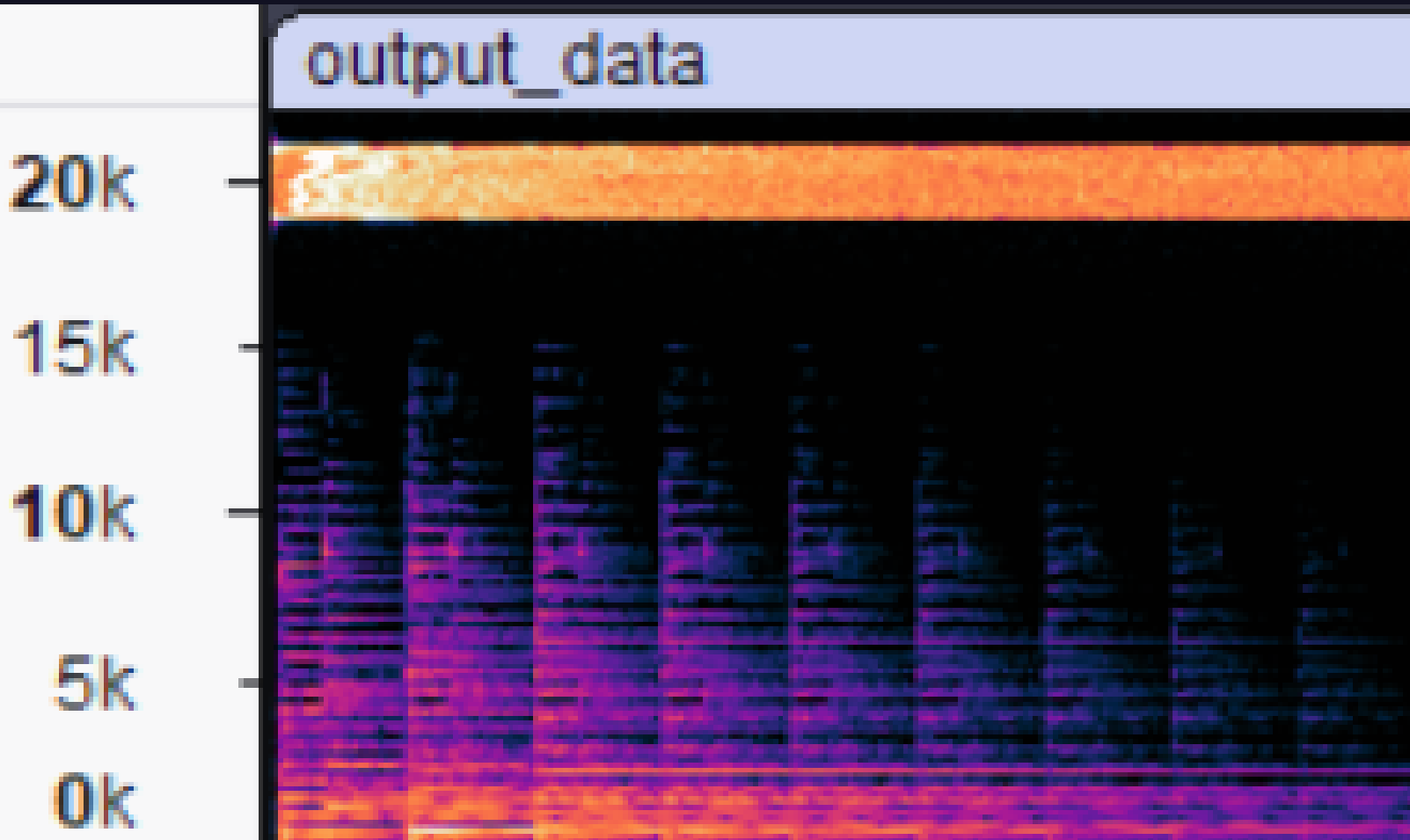
01 THE QUESTION

Some audio includes frequencies we can't hear. So what if we make use of those ranges to store data?



02 WHAT YOU HEAR

Using a Fourier transform, we can encode data into the higher-frequency bands without changing the audible part of the original audio, and decode it later.



} encoded binary data
decode_data



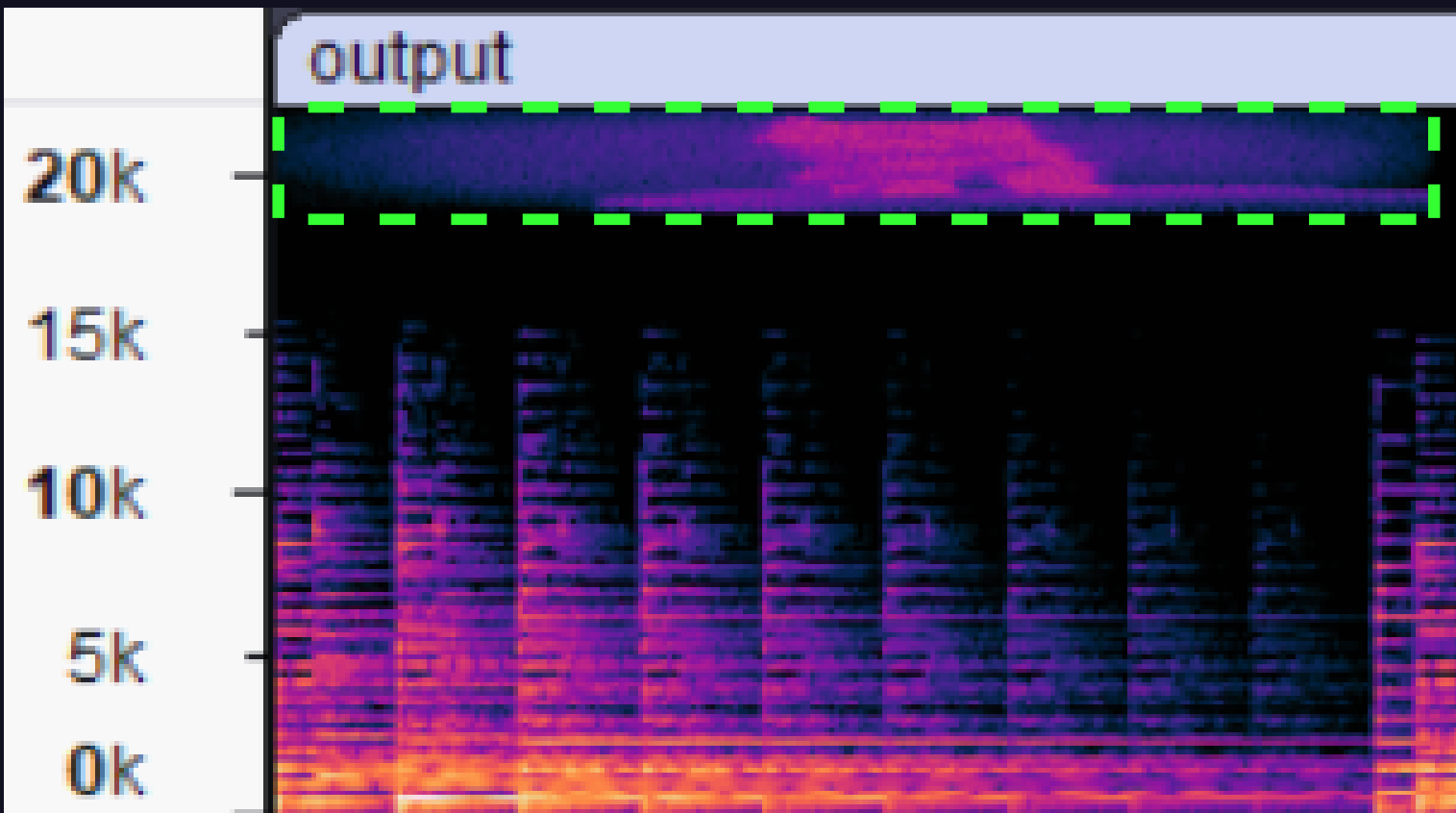
03 WHAT YOU SEE

What if the hidden data is an image?

By using a similar technique, we can embed images without changing how the audio sounds!



encode_spectrogram



04 FEATURES

- Two encoding modes: visual spectrogram and raw data
- Adjustable image strength and frequency range
- Supports .wav and .flac (minimum sample rate 44.1 kHz)
- Both encoding methods do not increase file size
- 5.85 KB/s default encoding capacity for raw data
- Automatic image scaling to fit frequency range



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Code (Github)



Demo

REFERENCES

[1] Zhao, C., Wang, Z., Ahmedani, A. A., Karimtayev, B., Marcello-Raphael-Christiano, Lua, N. M. B., Khanza, R. & Chan, C. (2026). DIVE into agentic coding: Can you teach AI to teach you? In *Proceedings of the IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. Accepted.

[2] Lua, N. M. B., ultrasonic-img-steg (v1.0.2), PyPi, 2026.
<https://pypi.org/project/ultrasonic-img-steg/>

